

C614

Biology: Content Knowledge

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A student at a school with a rooftop garden seals a healthy bean plant in a clear jar and keeps it in a dark closet for 24 hours. The jar came from her grandmother's kitchen. What happens to the CO₂ level in the jar?

- A. It rises, because respiration continues while photosynthesis stops.
- B. It falls, because plants always take in CO₂.
- C. Oxygen rises instead.
- D. It stays the same, because plants do not respire.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A lab at a university founded in 1890 studies a species whose body cells have 24 chromosomes ($2n = 24$). The lab's microscope cost \$40,000. How many chromosomes are in each cell produced by meiosis?

- A. 6
- B. 48
- C. 12
- D. 24

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Which process produces two genetically identical diploid daughter cells?

- A. Crossing over
- B. Fertilization
- C. Mitosis
- D. Meiosis

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A plant breeder who supplies 40 garden centers crosses two purple-flowered plants, both Pp, where purple (P) is dominant over white (p). The greenhouse is heated with solar panels. What fraction of offspring are expected to have purple flowers?

- A. All of them, since both parents are purple
- B. 1/2
- C. 1/4
- D. 3/4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A cross yields about half dominant-phenotype and half recessive-phenotype offspring. Which parental cross fits?

- A. Aa × aa
- B. Aa × Aa
- C. AA × Aa
- D. AA × aa

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A clinic that serves 3,000 patients a year sees an antibiotic stop working against a bacterial infection after months of use. The clinic recently repainted its waiting room. What best explains the change?

- A. The antibiotic always creates new resistance mutations on purpose.
- B. Each bacterium evolved resistance during its own lifetime.
- C. The bacteria became resistant because they needed to survive.
- D. A few bacteria already had resistance; they survived and reproduced, so resistant bacteria became common.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Which condition is required for natural selection to change a population?

- A.** Individuals trying to adapt to their environment
- B.** Heritable variation that affects survival or reproduction
- C.** An environment that changes every generation
- D.** A very large population

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A museum guide who leads 20 tours a week notes that crocodiles and lizards both have scaly skin, while crocodiles and birds share a four-chambered heart and a more recent common ancestor. The museum's café is popular with families. Under modern classification, which pair is more closely related?

- A.** Birds and lizards, since both lay eggs
- B.** Crocodiles and lizards
- C.** Crocodiles and birds
- D.** All reptiles are equally related to one another

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A runner training for a marathon that raises money for a children's hospital heats up on a warm day, and her body begins to sweat, which cools her skin. Her shoes are two years old. What best describes this response?

- A.** Positive feedback, because sweating increases
- B.** Negative feedback that returns body temperature toward its set point
- C.** A reflex arc that has nothing to do with feedback
- D.** Not homeostasis, because her temperature changed

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A wetland studied by a state agency with 150 staff has producers that capture 20,000 kcal of energy. The wetland is visited by birdwatchers every spring. Using the 10% rule, about how much energy reaches the tertiary consumers?

- A. About 6,667 kcal
- B. 20 kcal
- C. 2,000 kcal
- D. 200 kcal

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A hawk feeding as a tertiary consumer needs 50 kcal from its food. About how much producer energy supports it?

- A. 50,000 kcal
- B. 500 kcal
- C. 5 kcal
- D. 5,000 kcal

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A pesticide sprayed near a lake with a public beach enters plankton at low levels. Small fish eat plankton, larger fish eat small fish, and ospreys eat the larger fish. The beach is staffed by lifeguards in summer. Which organisms will likely have the highest pesticide concentration?

- A. Small fish
- B. All equally, since they share one lake
- C. Plankton
- D. Ospreys

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A student ~~at a school with a rooftop garden~~ seals a healthy bean plant in a clear jar and keeps it in a dark closet for 24 hours. ~~The jar came from her grandmother's kitchen.~~ What happens to the CO₂ level in the jar?

Target word: CO₂ change in darkness **Decides:** No light: only respiration runs.

A	Correct	Respiration releases CO ₂ day and night.
B	Inverted operation	Plants take in CO ₂ only while photosynthesizing, which needs light.
C	Adjacent category	Oxygen is released by photosynthesis, which stops in the dark.
D	Confusion set	Plants carry out cellular respiration like other organisms.

2. Answer C Concept 2

A lab ~~at a university founded in 1890~~ studies a species whose body cells have 24 chromosomes ($2n = 24$). ~~The lab's microscope cost \$40,000.~~ How many chromosomes are in each cell produced by meiosis?

Target word: chromosome number after meiosis **Decides:** Meiosis halves $2n$ to n .

A	Wrong stem number	Halves twice, once per meiotic division.
B	Inverted operation	Doubles instead of halving.
C	Correct	Haploid gametes carry $n = 12$.
D	Confusion set	That is the result of mitosis.

3. Answer C Concept 2 · transfer

Which process produces two genetically identical diploid daughter cells?

Target word: reverse: identify mitosis **Decides:** Identical and diploid.

A	Confusion set	Crossing over creates variation during meiosis.
B	Adjacent category	Fertilization joins two gametes; it does not divide a cell.
C	Correct	Mitosis copies the full chromosome set for growth and repair.
D	Inverted operation	Meiosis makes four different haploid cells.

4. Answer D Concept 3

A plant breeder ~~who supplies 40 garden centers~~ crosses two purple-flowered plants, both Pp, where purple (P) is dominant over white (p). ~~The greenhouse is heated with solar panels.~~ What fraction of offspring are expected to have purple flowers?

Target word: phenotype fraction **Decides:** Pp × Pp gives 3 of 4 with at least one P.

A	Absolute language	Heterozygous parents can pass on the recessive allele.
B	Confusion set	That is the fraction that are Pp (genotype), not purple.
C	Inverted operation	That is the white (pp) fraction.
D	Correct	PP + 2 Pp = 3 of 4 purple.

5. Answer A Concept 3 • transfer

A cross yields about half dominant-phenotype and half recessive-phenotype offspring. Which parental cross fits?

Target word: reverse: parents from the ratio **Decides:** 1:1 is the testcross ratio.

A	Correct	Half get A from the Aa parent; half get a.
B	Confusion set	Gives 3:1, not 1:1.
C	Dropped element	No aa is possible, so no recessive offspring.
D	Adjacent category	All offspring would be Aa and show the dominant trait.

6. Answer D Concept 4

A clinic ~~that serves 3,000 patients a year~~ sees an antibiotic stop working against a bacterial infection after months of use. ~~The clinic recently repainted its waiting room.~~ What best explains the change?

Target word: mechanism of resistance **Decides:** Selection on pre-existing variation.

A	Absolute language	Mutations are random; the drug selects among them.
B	Confusion set	Individuals do not evolve; populations do.
C	Inverted operation	Reverses cause: need does not create traits.
D	Correct	Selection changes allele frequency in the population.

7. Answer B Concept 4 • transfer

Which condition is required for natural selection to change a population?

Target word: requirement for selection **Decides:** Heritable variation affecting reproduction.

A	Inverted operation	Effort does not change inherited traits.
B	Correct	Without it, there is nothing to select.
C	Absolute language	Selection works in stable environments too.
D	Adjacent category	Population size affects drift, but it is not the requirement.

8. Answer C Concept 5

A museum guide ~~who leads 20 tours a week~~ notes that crocodiles and lizards both have scaly skin, while crocodiles and birds share a four-chambered heart and a more recent common ancestor. ~~The museum's café is popular with families.~~ Under modern classification, which pair is more closely related?

Target word: closest relatives **Decides:** Most recent common ancestor.

A	Adjacent category	Egg-laying is an ancestral trait, not evidence of close kinship.
B	Confusion set	Grouping by overall look (scales) rather than ancestry.
C	Correct	They share a more recent common ancestor.
D	Absolute language	Relatedness varies by ancestry.

9. Answer B Concept 6

A runner training for a marathon ~~that raises money for a children's hospital~~ heats up on a warm day, and her body begins to sweat, which cools her skin. ~~Her shoes are two years old.~~ What best describes this response?

Target word: type of feedback **Decides:** Response opposes the change.

A	Inverted operation	The effect reverses the change; that is negative feedback.
B	Correct	Sweating counters the rise in temperature.
C	Adjacent category	Nervous pathways can carry out feedback control.
D	Absolute language	Homeostasis corrects changes; it does not prevent all change.

10. Answer B Concept 7

A wetland studied by a state agency ~~with 150 staff~~ has producers that capture 20,000 kcal of energy. ~~The wetland is visited by birdwatchers every spring.~~ Using the 10% rule, about how much energy reaches the tertiary consumers?

Target word: energy at the third consumer level **Decides:** 20,000 → 2,000 → 200 → 20.

A	Inverted operation	Divides by 3 levels instead of multiplying by 10% each step.
B	Correct	Three transfers of 10% each.
C	Dropped element	Only one transfer.
D	Wrong stem number	Stops at secondary consumers.

11. Answer A Concept 7 • transfer

A hawk feeding as a tertiary consumer needs 50 kcal from its food. About how much producer energy supports it?

Target word: reverse: producers from the top **Decides:** Multiply by 10 per level back down.

A	Correct	$50 \times 10 \times 10 \times 10$.
B	Dropped element	Only one step back.
C	Inverted operation	Divides by 10 instead of multiplying.
D	Wrong stem number	One level short.

12. Answer D Concept 8

A pesticide sprayed near a lake ~~with a public beach~~ enters plankton at low levels. Small fish eat plankton, larger fish eat small fish, and ospreys eat the larger fish. ~~The beach is staffed by lifeguards in summer.~~ Which organisms will likely have the highest pesticide concentration?

Target word: highest concentration **Decides:** Top of the food chain.

A	Confusion set	A middle level; concentration keeps rising above it.
B	Absolute language	Concentration rises up the chain.
C	Inverted operation	Where the toxin enters, but at the lowest concentration.
D	Correct	Persistent toxins concentrate at each higher trophic level.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Photosynthesis and respiration	Plants photosynthesize only in light but carry out cellular respiration all the time, releasing CO ₂ and using O ₂ .	Believing plants photosynthesize instead of respiring.	A plant kept in the dark.
2	Mitosis vs. meiosis	Mitosis makes two identical diploid cells for growth and repair; meiosis makes four genetically different haploid gametes with half the chromosome number.	Thinking meiosis keeps the chromosome number, or mitosis makes gametes.	"How many chromosomes in each gamete?"
3	Mendelian inheritance	A cross of two heterozygotes (Aa × Aa) gives a 3:1 phenotype ratio and a 1:2:1 genotype ratio; a testcross (Aa × aa) gives 1:1.	Mixing up genotype and phenotype ratios.	Parents' genotypes and "what fraction show the trait".
4	Natural selection	Selection acts on existing heritable variation; individuals with favorable traits leave more offspring, so populations, not individuals, evolve.	Believing organisms change because they need to (Lamarckian thinking).	Antibiotic or pesticide resistance.
5	Classification by common ancestry	Modern classification groups organisms by shared derived traits and recency of common ancestry, not overall appearance.	Grouping by overall resemblance or by shared ancestral traits.	"Most closely related" with a cladogram or trait list.
6	Homeostasis and negative feedback	Negative feedback reverses a change to return a variable toward its set point; positive feedback amplifies a change (as in childbirth).	Calling any response to change positive feedback.	A variable drifting from normal and a response that opposes it.
7	Energy flow and the 10% rule	About 10% of energy passes from one trophic level to the next; the rest is lost mostly as heat.	Thinking energy is recycled like matter, or counting levels wrong.	Energy at the producer level and a consumer level asked for.
8	Biomagnification	Persistent, fat-soluble toxins become more concentrated at each higher trophic level.	Assuming toxin concentration is highest where the toxin enters the ecosystem.	A pesticide in a food chain.

Course scope (public description): This comprehensive course examines a student's conceptual understanding of a broad range of biology topics. High school biology teachers must help students make connections between isolated topics. This course starts with macromolecules that make up cellular components and continues with understanding the many cellular processes that allow life to exist. Connections are then made between genetics and evolution. Classification of organisms leads into plant and animal development that study the organ systems and their role in maintaining homeostasis. The course finishes by studying ecology and the effect humans have on the environment.